Final Project Report

Course Number: CYB 6500 CAP1.

Course Title: Cybersecurity Capstone.

1. **Analysis of the initial problem**

Organizations are struggling to keep up with sophisticated threats like Advanced persistent threats (APTs), ransomware, and nation-state tactics, against which traditional security often reacts too late, triggering alerts only post-damage. Existing honeypot solutions frequently offer insufficient insight into attacker methodologies, leaving critical sectors like finance, defense, and healthcare susceptible to targeted attacks without understanding adversary intent (CrowdStrike, 2023).

Additionally, many honeypots are prone to detection via tools such as Shodan or protocol-level scans, diminishing their effectiveness once identified. Our project offers a solution to a stealth-capable honeypot system, including Dionaea, Cowrie, Honeyd, OpenCanary, Conpot, etc., which are deployed and designed to resist detection through anti-fingerprinting techniques and realistic service emulation. This system, integrated with attack path reconstruction, MITRE ATT&CK mapping, and a government-exclusive backtracking module, aims to provide early-warning intelligence, enhance defensive readiness, and support attribution within governmental frameworks (Morić, Z., Dakić, V., & Regvart, D., 2025).

2. **Discussion of improvement opportunities**

The Smart Honeypot Network is designed to enhance cybersecurity defenses by creating an intelligent, deceptive environment for attackers. The project focuses on deploying undetectable honeypots that mimic real-world services to attract and analyze malicious activities within an isolated virtual environment. This approach aims to provide proactive insights into attacker tactics, techniques, and procedures rather than merely reacting to breaches.

A key aspect of this system is its ability to capture attacker behavior and reconstruct their attack paths, providing a visual timeline of events. This information is then used to generate actionable threat intelligence reports, which are aligned with the MITRE ATT&CK framework, offering a standardized understanding of adversary actions. The project also plans to develop an analyst dashboard that will centralize this intelligence, making it easier for security teams to monitor and respond to threats effectively.

Additionally, a distinctive feature of the Smart Honeypot system is the proposal for a government-exclusive backtracking module. This module is intended to support attribution efforts and enable controlled counter-engagement against attackers. By resisting detection through anti-fingerprinting techniques and realistic service emulation, the honeypot network aims to be stealth-capable, making it more effective in luring and gathering intelligence from sophisticated adversaries who often use tools like Shodan to identify and avoid traditional honeypots.

3. **Business case for identified improvements**

The Smart Honeypot System presents a robust business case by transforming traditional, reactive defenses into proactive threat intelligence platforms. By deploying intelligent honeypots that mimic real-world services and actively lure attackers into isolated environments, organizations can gain invaluable early-warning intelligence. By taking action early, the period of inactivity or unresponsiveness is shortened, which can reduce breach remediation costs by an estimated 30–40%, and plays a vital role in mitigating the substantial financial and reputational damage caused by cyberattacks, as breaches detected later in their lifecycle incur significantly higher remediation costs (Verma, A. S., & Dubey, A. 2020).

Also, the system's ability to reconstruct attack paths and map observed behaviors to the MITRE ATT&CK framework directly contributes to improved incident response readiness and the refinement of existing defensive playbooks. This allows security teams to prioritize mitigations based on real-world adversary tactics, rather than generic vulnerabilities, leading to more efficient allocation of cybersecurity resources. The government-exclusive backtracking module, while restricted, proposes an advanced capability for attribution, which can deter future attacks and support legal or operational actions when authorized, further safeguarding national and critical infrastructure assets.

The investment in such a system is justified by its potential to deliver a high return on investment (ROI) through enhanced defensive capabilities and reduced incident costs. By enabling early detection, deeper insight into adversary intent, and supporting attribution efforts, the Smart Honeypot System directly addresses the limitations of traditional reactive defenses and the vulnerability of high-value sectors to targeted attacks. This strategic shift from merely responding to breaches to proactively understanding and preventing them underscores the compelling business rationale for adopting this advanced deception technology.

4. **Project plan mapping out the scope, key deliverables, and a timeline for the suggested changes**

Project Scope

The project is centered around setting up a comprehensive honeypot network, which involves extensive research into diverse honeypot technologies and the strategic design of network topology within an isolated lab environment, followed by the deployment and optimization of various honeypot services like SSH, SMB, web, and databases, alongside robust logging and monitoring infrastructure. Concurrently, the project focuses on developing an advanced analytical tool to reconstruct attack paths, visualize attacker timelines, and generate automated threat intelligence reports mapped to frameworks such as MITRE ATT&CK, ultimately leading to the creation of an analyst dashboard for enhanced security operations. Furthermore, a critical component of this project scope is the proposal for a specialized, government-exclusive backtracking module, designed to facilitate attribution and controlled counter-engagement, thereby contributing significantly to advanced defensive capabilities.

Key Deliverables

- Live Honeypot Network Demonstration: This deliverable showcases a deployed, fully operational intelligent honeypot network, showcasing its ability to lure and capture attacker traffic within a secure and isolated environment while resisting detection. It integrates various honeypot technologies, including at least Cowrie and Dionaea, while actively resisting detection.
- Attack Path Visualization Dashboard: A dynamic dashboard powered by tools like Neo4j for effective correlation will be provided, graphically representing the sequence of attacker actions within the honeypot environment to enable a clear understanding of attack progressions.
- Threat Intelligence Reports from Captured Attacker Activity: Detailed, actionable threat intelligence reports will be generated from captured attacker behaviors, mapping TTPs to frameworks like MITRE ATT&CK for early warning and proactive defense.
- Return on Investment (ROI) Analysis Showing Early Detection Value: A comprehensive analysis will demonstrate the economic benefits of early threat detection through the honeypot system, quantifying reduced incident costs and improved security posture.
- Future Work Outline for Government-Grade Backtracking Module: A structured outline will detail the proposed development of a specialized backtracking module for government contexts, focusing on attribution and controlled counter-engagement capabilities.

- Deployment Guide and Final Capstone Report: This includes a step-by-step deployment guide for the honeypot network and a comprehensive capstone report documenting the project's objectives, methodology, outcomes, and lessons learned.

Timeline

The project spanned a four-month period, during which a comprehensive network of honeypots was researched, deployed, and continuously optimized. This systematic approach ensured that each phase built upon the previous one, leading to a robust and intelligent honeypot system.

The first month was dedicated to foundational research and setting up the environment. This included an extensive study of various honeypot technologies like Cowrie, Dionaea, OpenCanary, Honeyd, Conpot, and T-Pot to determine the best fit for mimicking real-world services. This also included a very specific network topology design to ensure proper isolation and efficient data flow within the honeypot environment. At the end of this phase, an isolated lab environment equipped with multiple foundational honeypots was set up and prepared for further deployment.

The second month was about deploying different honeypot services and implementing extensive logging and monitoring. During this phase, services such as web servers, SSH, SMB, and databases were deployed in the isolated lab, each configured to emulate legitimate systems. An extensive logging and monitoring infrastructure, leveraging tools like Zeek (Bro), tcpdump, tshark, Suricata, Snort, nfcapd/nfdump, Elasticsearch, Logstash, Filebeat/Packetbeat, and Kibana, was also established to capture all interactions and activities for in-depth analysis.

This third month was important for advanced analytics development. Much emphasis would be laid on creating sophisticated attack path visualization tools, such as Neo4j, NetworkX, Cytoscape, and Gephi that could allow the graphical representation of the movements and actions of the attackers in the honeypot environment.

The final month was dedicated to refining the system, presenting the findings, and documenting the entire project. This involved the design and implementation of a user-friendly analyst dashboard, specifically tailored for reconstructing attack timelines and providing a clear overview of security incidents. Detection algorithms were optimized to enhance their accuracy and efficiency.

As a forward-looking component, a simulated backtracking workflow was also presented, highlighting its potential as a restricted government-use extension for attribution and counter-engagement, though its actual implementation was beyond the scope of this project.

5. **Project tracking and success metrics**

Deliverable	**Tracking Indicator**	**Success Metric**
Functional Honeypot Deployment	Live traffic capture and honeypot service availability	100% operational status of deployed honeypots (SSH, Web, SMB, DB) with captured attacker interactions.
Attack Path Visualization Dashboard	Number of distinct attack paths visualized	Visualization of at least three distinct, reconstructed attack paths with timeline details.
Automated MITRE ATT&CK Aligned Reports	Generation of intelligence reports with ATT&CK mapping	Production of automated threat intelligence reports that accurately map observed attacker behaviors to MITRE ATT&CK techniques.
Analyst Dashboard Demonstrating Timeline Reconstruction	Dashboard functionality and reconstruction accuracy	Fully functional analyst dashboard capable of reconstructing attack timelines and displaying relevant data.
Documented and fully operational Proposal for Government Backtracking module	Completion and detail of the backtracking module proposal	Comprehensive proposal for a government-exclusive backtracking module, outlining its conceptual workflow and legal/ethical considerations.
Documentation and Presentation of the Project	Completed Project	Final Completed Project delivered within a timeline of 4 months.

6. **Summary of project outcomes and lessons learned**

The Smart Honeypot project successfully achieved its objectives by building and evaluating a sophisticated honeynet environment. Key outcomes included the deployment of diverse honeypots

mimicking real services, the effective capture and analysis of attacker behaviors using the ELK stack and Neo4j, and the generation of actionable threat intelligence reports aligned with MITRE ATT&CK. The project demonstrated the feasibility of reconstructing attack paths with timeline visualization and presented a functional analyst dashboard, thereby enhancing early warning capabilities and providing deeper insights into adversary tactics.

Throughout the project, several valuable lessons were learned, primarily concerning the iterative nature of anti-fingerprinting techniques. It became clear that continuous tuning and adaptation are essential to maintain the realism and effectiveness of deception environments against evolving attacker methodologies. Challenges related to managing large volumes of data for logging and retention, as well as the complexity of correlating attack sessions across disparate honeypots, highlighted the need for robust data models and careful planning.

The project also underscored the critical importance of addressing legal and ethical considerations, especially for sensitive components like the proposed government-exclusive backtracking module. Maintaining strict isolation and egress controls was paramount to ensuring the operational safety of the honeypot environment and preventing misuse of captured malware. These insights provide a strong foundation for future enhancements and highlight areas requiring careful consideration in real-world deployments.

References:

CrowdStrike. (2023). CrowdStrike 2023 Global Threat Report – From Relentless Adversaries to Resilient Businesses. Retrieved from https://tinyurl.com/mrydcuub

Morić, Z., Dakić, V., & Regvart, D. (2025). Advancing Cybersecurity with Honeypots and Deception Strategies. In *Informatics* (Vol. 12, No. 1, p. 14). MDPI AG.

Verma, A. S., & Dubey, A. (2020). A Review on Honeypot Deployment. *LJP London Journal of Research in Computer Science and Technology*, *20*(1).

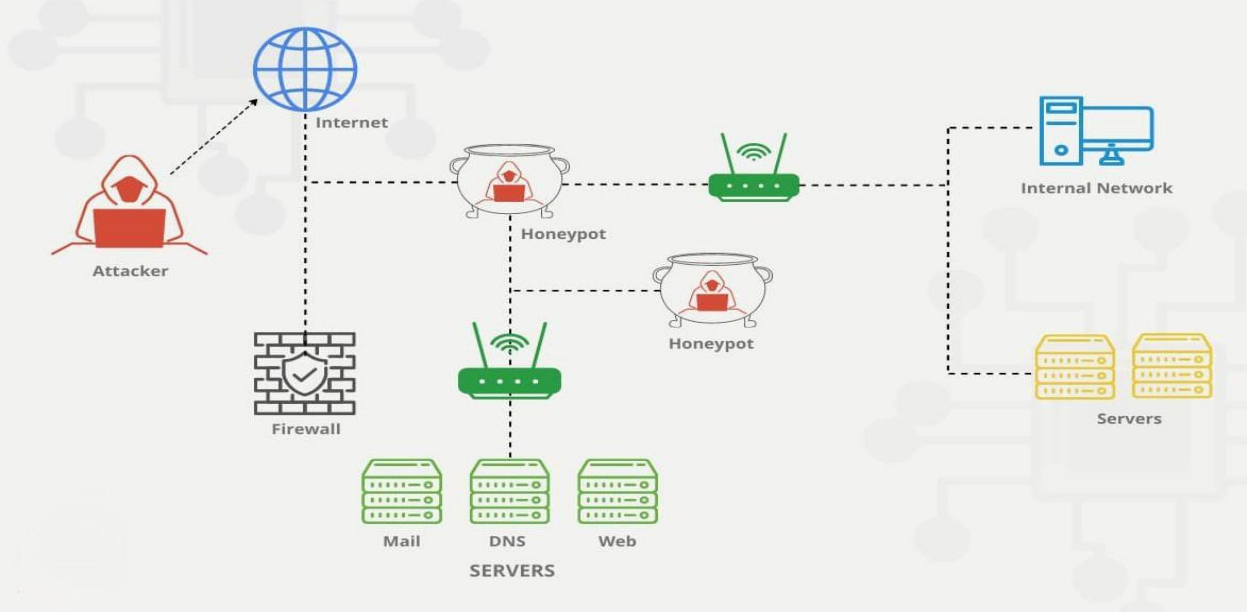
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Appendices

Appendix A — Project Team and Contributions

- **Syed Abdul Malik:** Lead architect, responsible for honeypot deployment, ELK stack configuration, and Backtracking module design.
- **Hamuza Senyonga:** Documentation, Neo4j data modeling, and attack path correlation dashboard/visualization development.

Appendix B: Design and Architecture



Network and Isolation

- Virtualized lab using nested virtualization (VM hosts and dedicated VLANs) to ensure honeypots are isolated from production networks. All outbound traffic from honeypots is routed through a controlled egress proxy to prevent misuse.
- Segmented topology: frontend deception VLAN for low-interaction services, high-interaction VLAN for deeper engagement, and monitoring VLAN for ELK/Neo4j and admin access.

Honeypot Stack

- Low- and high-interaction mix:
    - Cowrie (SSH/Telnet) — customized to mimic common SSH banners and shell artifacts.
    - Dionaea — captures malware payloads and SMB interactions.
    - Custom web application honeypot — deliberately vulnerable endpoints and dynamic content generation to appear realistic.
    - Lightweight database emulator — accepts connections and logs queries.
- Containerized deployment for rapid reset and scaling; orchestration via docker-compose for the prototype with notes for Kubernetes for production scaling.

Anti-Fingerprinting Measures

- Service banner variability: randomized but plausible service strings and OS fingerprints to avoid deterministic detection.
- Timing variability and simulated jitter: introduce realistic network latency and inter-packet timing to mimic genuine hosts.
- Emulation of filesystem artifacts and user accounts: decoy files, realistic directory structure, and plausible user accounts with last-login timestamps.
- Dynamic content generation: web honeypot generates pages, error traces, and “user activity” logs to confuse basic scanners.
- Logging of probing fingerprints to refine anti-fingerprinting (adaptive hardening).

Logging, Analysis, and Storage

- Log ingestion: Logstash pipelines normalize logs from honeypots into JSON with standardized fields (timestamp, srcip, dstservice, payloadhash, sessionid, captured_command).
- Storage and search: Elasticsearch indexes for fast querying; retention policy configurable (prototype uses 90 days).
- Visualization: Kibana dashboards for high-level metrics (top attackers, geolocation heatmap, service-based timelines).
- Evidence store: raw session captures (pcap and session transcripts) preserved with cryptographic hashes for integrity.

Attack Path Correlation (Neo4j)

- Data model:
    - Nodes: Attacker (IP/session), Host (honeypot instance), Service, Action (e.g., exploit, command execution), Artifact (file/hash), Timepoint.
    - Edges: performedby, targeted, resultedin, temporalfollowedby.
- Path reconstruction algorithm: temporal ordering by session timestamps, heuristic correlation by shared artifacts (uploaded files, identical payload hashes), and action sequence linking.
- Visualization: Neo4j Bloom / custom D3 frontend to render lateral movement and step-by-step sequences as timelines.

MITRE ATT&CK Mapping and Report Generation

- Behavioral engine maps observed actions and extracted indicators (syscalls, commands, persistence attempts, C2 patterns) to ATT&CK technique IDs.
- Automated report templates include: summary, timeline, ATT&CK matrix with highlighted techniques, IOC table, recommended mitigations, and evidence links.
- Export formats: PDF for executive/distribution, JSON for SOC ingestion, STIX/TAXII compatible payload for threat sharing.

ELK stack and Neo4j

- ELK Stack (Elasticsearch, Logstash, Kibana): Primarily an observability and search solution, excellent for handling unstructured or semi-structured log, time-series, and document data for full-text search, aggregation, and real-time monitoring.

Focus: Indexing, searching, and visualizing documents.

- Neo4j: A native Graph Database optimized for storing data where the relationships are as important as the data points (nodes) themselves. It's built for rapid traversal of connections.

Appendix C: Simulation

https://drive.google.com/drive/folders/1AZ3HjO2chZWXOVlCmV4DfI_r7EgL1_x_?usp=drive_link